Case study: How does a bike-share navigate speedy success?

INTRODUCTION

Cyclistic is a bike share company in Chicago. The company provides membership and casual rides services with more than 5800 geo-tracked bicycles across 692 stations throughout Chicago. It have been in operation since 2016 and wants to improve even more the service.
To be more inclusive, Cyclistic also offers tricycles, reclining and cargo bikes. About 8% of the rides are with the assistive options, the rest is the traditional.
Cyclistics users are more likely to ride for leisure, but about 30% use the bikes to commute to work each day.

PROBLEMS

So far, Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders and Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments.

BUSINESS TASK

How do annual members and casual riders use Cyclistic bikes differently?

The team needs to better understand how annual members and casual riders differ in order to create marketing campaigns to convert casual riders into members.

DATA SOURCES

The company has a public database of the rides. The data is published monthly and for reason of data privacy, data that contains information that can be used to identify users have been removed. So only informations that can be useful to analyse trends about the rides are available.

CLEANING AND MANIPULATION OF DATA

The data were checked for any incongruence on data format within the columns and rows, and if the names were typed correctly and if they were matching with the other files. After, using the statistical program R, the data were compiled in one, and more data cleaning were processed, such as removing ride entries with rides with negative duration and considering the rides with more than 1 minute of duration.

Another column were added to calculate the average time of each ride to easily help identify trends between the casual and members rides.

It was analysed the number of rides per weekday and per month, and it also analysed the the average duration of the rides.

SUMMARY OF ANALYSIS

For the year of 2021 were registered more than 5.5M rides. Casual rides had more than 2.4M rides and 3M rides for members, the proportion of the rides can be seen on Figure 1 (pie chart). The average ride duration of casual riders were 32 minutes and members were 14 minutes.

The year of 2022 Cyclistic registered 5.54M rides, a few more rides registered than 2021. It was registered 2.2M casual rides and 3.2M member rides, the proportion of rides can be seen on the Figure 1. The average ride duration of casual riders were 29.8 minutes and members were 13 minutes.

For the year of 2023, there were 5.56M rides, more than the previous years. The number of casual rides for this year was around 2M rides and 3.6M of member rides, the proportion of rides can be seen on the Figure 1. The average duration for casual rides were 29 minutes and for member rides were 13 minutes.

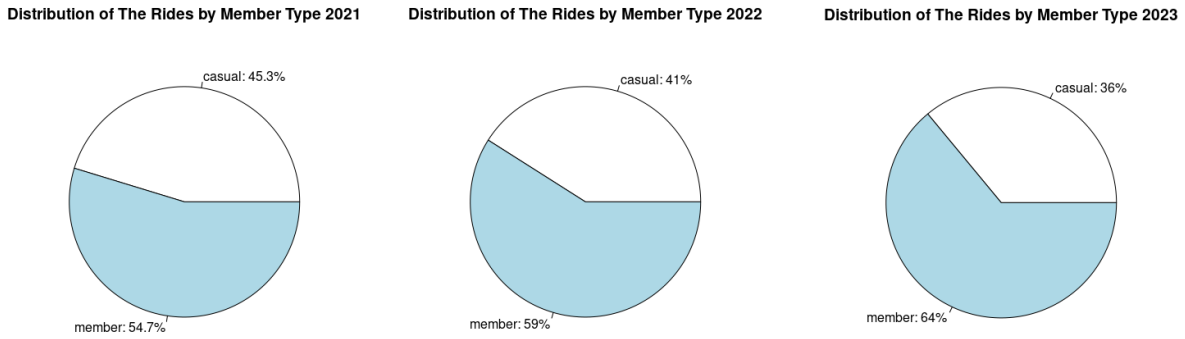

Figure 1. Distribution of the rides by year.

Along the 3 years it was possible to notice trends that helps to understand the use of casual and member rides and how they differ. It is possible to see on Figure 2 and 3, that the member rides are shorter and maintain the same average throughout the year. The casual rides are longer and increase during the weekend and during the months with the hottest temperatures.

Figure 2: Average ride duration on the weekdays by year.

Figure 3: Average ride duration each month by year.

Related to the number of rides, it is possible to see on Figures 4 and 5, the during the weekdays, members have a larger number of rides and it decreases during weekend, the opposite happens with casual rides. And for this analysis it can also be observed an increase of number of rides during the months with the hottest temperatures.

Figure 4: Number of rides during the weekdays.

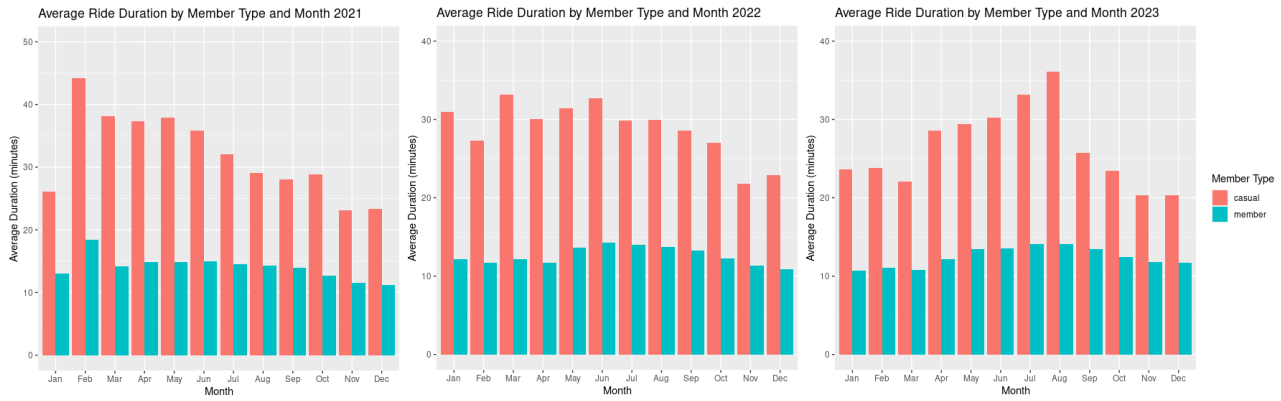

Figure 5: Number of rides each month by year.

CONCLUSIONS (Key findings)

With the 3 years of data used, it was possible to notice an increase of number of rides in total and an increase of member rides every year. The increase of rides happens during the hottest months, what can clearly shows that users are trying to spend more time outdoors.

The average duration shows that members and casual riders differ with member having a shorter ride duration possibly indicating a more planned and direct ride to go to work or home, the casual rides are possibly exploring the city. This can also explain the increase of the number of rides during the weekend by casual members.

SOLUTIONS

In order to attract more casual riders to acquire the annual membership, marketing campaign targeting only casual riders during the hottest months could attract this riders with the idea they would be saving money on each ride they do if they do the adhesion.

NEXT STEPS (Top three recommendations)

1. Offer a package with some discounted rides in order to attract more casual rides

2. More propaganda on the street and through push notifications on the app reinforcing the idea that the bicycle can be used as a means of transportation for work.

3. Invite a friend and get a discount on the next annual membership renewal.